

REMA

RAPID EMERGENCY MEDICAL ACCESS

OPERATING PROTOCOL

Field Reference Document — All Phases

Organizer	Viet Nam Red Cross
Target City	Ho Chi Minh City (or comparable Vietnamese delta city)
Scenario	Recurrent urban flood — 5 to 7 days, 30 to 80 cm water depth
Critical Window	Hours 0 to 48
Document Version	v1.0 — University Track Submission
Classification	FOR OFFICIAL USE — Red Cross Staff and Named Volunteers Only

■ CRITICAL SAFETY RULE: All delivery operations are SUSPENDED when water depth exceeds 80 cm. Escalate immediately to civil defense. No exceptions.

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SECTION 1 — PHASE 1 ACTIVATION CHECKLIST

Hours 0 to 24

ACTIVATION TRIGGER: Phase 1 activates when ANY 2 of these 3 conditions are met simultaneously: (1) City/provincial flood warning Level 2 or above | (2) Rainfall forecast exceeds 100 mm in 24 hours | (3) Any target district reports street-level flooding

HOURS 0–3 — Alert and Mobilize

Decision-maker: Emergency Coordinator (Operations Center)

	Action	Responsible
■	Confirm trigger conditions — record which 2 of 3 are met	Emergency Coordinator
■	Issue formal REMA activation order (phone + SMS to all Hub Managers)	Emergency Coordinator
■	Notify Hub Managers: District 1, District 2, District 3 on standby	Emergency Coordinator
■	Alert volunteer team leaders in each district (via Hub Manager)	Hub Managers (×3)
■	Operations Center: pull pre-packed EMKs from central warehouse storage	Warehouse Manager
■	Confirm Ministry of Health (MoH) contact — request EMK-3 cold chain release	Emergency Coordinator
■	Begin REMA dashboard tracking — set phase to Phase 1	Operations Staff
■	Check radio equipment — confirm all 4 handsets charged and on agreed frequency	Operations Staff
■	Record activation time: _____ Recorded by: _____	—

HOURS 3–8 — Load and Dispatch to Sub-Warehouses

Decision-maker: Warehouse Manager + Emergency Coordinator for reallocation

	Action	Responsible
■	Load pre-packed EMK pallets onto trucks (3 trucks minimum)	Warehouse Manager
■	Verify dispatch quantities against printed manifest (EMK-1 / EMK-2 / EMK-3)	Warehouse Manager
■	Dispatch convoy simultaneously to all 3 sub-warehouse sites	Warehouse Manager
■	Use highest-elevation road routes — avoid roads below 1.5 m elevation	Truck drivers
■	Hub Manager District 1: receive stock, verify count, sign manifest	Hub Manager D1
■	Hub Manager District 2: receive stock, verify count, sign manifest	Hub Manager D2
■	Hub Manager District 3: receive stock, verify count, sign manifest	Hub Manager D3
■	5% spot-check: open boxes, verify contents match manifest — reject damaged stock	Hub Managers
■	Confirm EMK-3 cold box received from MoH: seal intact, temperature log shows <8°C	Hub Managers
■	Sub-warehouse setup: raise pallets, label zones (EMK-1 / EMK-2 / EMK-3), post zone map	Hub Managers
■	Set up volunteer check-in table at sub-warehouse entrance	Hub Managers
■	Confirm radio/phone contact with Operations Center from each sub-warehouse	Hub Managers
■	30% reserve confirmed at central warehouse — do NOT dispatch	Warehouse Manager

HOURS 8–16 — Volunteer Deployment and Community Assessment

Decision-maker: Hub Managers (district level) — report to Operations Center every 4 hours

	Action	Responsible
■	Deploy volunteer teams on foot or motorbike for assessment (NOT delivery yet)	Hub Managers
■	Each team assigned to pre-mapped zone — confirm zone assignment in writing	Hub Managers
■	Volunteers complete paper household assessment forms (~3 min per household)	Volunteer Teams
■	Collect: household size, elderly/pregnant/disabled, chronic illness, medication status	Volunteer Teams
■	Teams return to sub-warehouse every 4 hours to submit forms	Volunteer Teams
■	Hub Manager compiles forms — first demand estimate sent to Operations Center (WhatsApp/SMS)	Hub Managers
■	Report any route blockages found during assessment runs immediately	Team Leaders
■	Log route water depths by zone for routing decisions	Hub Managers
■	First radio check-in at 08:00 if already past that time, or set for 12:00	Hub Managers

HOURS 16–24 — First Demand Picture and Reallocation Decision

Decision-maker: Emergency Coordinator — reallocation window before roads deteriorate further

	Action	Responsible
■	Operations Center compiles all district demand reports into consolidated estimate	Operations Staff
■	Emergency Coordinator reviews: does any sub-warehouse need emergency resupply before Hour 24?	Emergency Coordinator
■	If YES — dispatch emergency resupply convoy from central warehouse reserve before roads close	Emergency Coordinator
■	Begin first delivery runs from sub-warehouses — CRITICAL band (15–20 pts) households only at this stage	Hub Managers
■	Confirm delivery mode per zone matches current water depth (see delivery mode table)	Hub Managers
■	Update REMA dashboard with Phase 1 demand summary and resupply status	Operations Staff
■	Prepare prioritized household list for Phase 2 — sorted by score band (Critical > High > Medium > Standard)	Hub Managers
■	Phase 1 complete — operations continue into Phase 2 at Hour 24	Emergency Coordinator

SECTION 2 — PHASE 2 ADAPTIVE DELIVERY CHECKLIST

Hours 24 to 48

DELIVERY MODE TABLE — Check Water Depth Before Each Run

Water Depth	Delivery Mode	Carry Capacity	EMKs per Trip
0 – 30 cm	Motorbike with waterproof pannier bags	~25 kg	5× EMK-1 or 3× EMK-2
30 – 60 cm	Cargo bicycle or volunteer on foot with backpack	~10–20 kg	2–4× EMK-1 or 1–2× EMK-2
60 – 80 cm	Small motorized boat or inflatable raft	~150 kg	30× EMK-1 or 20× EMK-2
> 80 cm	DELIVERY SUSPENDED — Escalate to civil defense	—	—

DELIVERY PRIORITY ORDER

Band	Score	Action
[CRITICAL]	15–20 pts	Deliver in current run — goes first
[HIGH]	10–14 pts	Deliver same day
[MEDIUM]	5–9 pts	Deliver within 48 hours
[STANDARD]	0–4 pts	Community collection point self-pickup

PHASE 2 DAILY OPERATIONS CHECKLIST (Repeat Each Day)

	Action	Responsible
■	Confirm water depth per zone before dispatching teams — update delivery mode if needed	Hub Manager
■	Dispatch Team A (zone 1) at 07:00 — prioritized list confirmed	Hub Manager
■	Dispatch Team B (zone 2) at 07:00 — prioritized list confirmed	Hub Manager
■	Dispatch Team C (zone 3) at 07:00 — prioritized list confirmed	Hub Manager
■	08:00 Radio check-in — report stock, incidents, morning plan to Operations Center	Hub Manager
■	Teams return between runs to restock and report new critical cases found in field	Team Leaders
■	12:00 Radio check-in — delivery progress, new critical cases, route issues	Hub Manager
■	Stock check: if any EMK type falls below 30% of original allocation — report SCARCITY to EC immediately	Hub Manager
■	Second delivery dispatch at 11:00 — address new cases reported in morning	Hub Manager
■	16:00 Radio check-in — afternoon delivery summary, resupply needs	Hub Manager
■	Third delivery dispatch at 15:00 if stock and daylight allow	Hub Manager
■	If water depth in any zone exceeds 80 cm — suspend all delivery, escalate to civil defense	Hub Manager
■	All signed delivery receipts submitted by returning teams — logged in stock ledger	Hub Manager
■	20:00 Radio check-in — end-of-day stock count, next-day plan, overnight incidents	Hub Manager
■	Operations Center updates dashboard with district delivery totals	Operations Staff
■	Emergency Coordinator reviews: any cross-district reallocation needed before next morning?	Emergency Coordinator

SECTION 3 — RADIO CHECK-IN SCRIPT

4 Fixed Daily Slots

Use this script when phone/internet is unavailable. All Hub Managers report to Operations Center at exactly 08:00, 12:00, 16:00, and 20:00. Use agreed radio frequency. Keep transmissions under 3 minutes.

08:00 — MORNING CHECK-IN

HUB MANAGER CALLS:	Hub Manager: "Operations, this is Hub [District Number]. Morning check-in. Over."
OPS CENTER RESPONDS:	"Hub [District], Operations. Send your report. Over."
REPORT FORMAT (Hub Manager):	
1. STOCK STATUS:	"EMK-1: [remaining] units. EMK-2: [remaining] units. EMK-3: [remaining] units."
2. OVERNIGHT INCIDENTS:	"[No incidents / Incident: describe briefly]."
3. MORNING PLAN:	"Dispatching [number] teams to zones [list zones]. First run at 07:00."
4. ROUTE STATUS:	"Routes clear / Route blocked at [location] — using alternate route."
5. RESUPPLY NEEDED?	"No resupply needed / Request resupply of [EMK type, quantity]."
CLOSE:	"Hub [District] reporting complete. Over and out."

12:00 — MIDDAY CHECK-IN

HUB MANAGER CALLS:	"Operations, this is Hub [District Number]. Midday check-in. Over."
REPORT FORMAT:	
1. DELIVERY PROGRESS:	"[Number] households reached this morning. [Number] CRITICAL remaining."
2. NEW CRITICAL CASES:	"[Number] new critical cases identified in field. Addresses: [brief list]."
3. ROUTE ISSUES:	"[No issues / Route in zone [X] now impassable — requesting boat transfer]."
4. STOCK UPDATE:	"Stock now at [%] of original allocation. [Normal / Approaching scarcity threshold]."
5. AFTERNOON PLAN:	"Dispatching second run at 11:00 to [zones]. Estimated [number] households."

16:00 — AFTERNOON CHECK-IN

HUB MANAGER CALLS:	"Operations, this is Hub [District Number]. Afternoon check-in. Over."
REPORT FORMAT:	
1. DELIVERY TOTAL TODAY:	"[Number] households reached total today."
2. REMAINING PRIORITY:	"[Number] HIGH remaining. [Number] MEDIUM remaining."
3. RESUPPLY REQUEST:	"[No resupply / Request: EMK-[type], [quantity], needed by [time]]."
4. SAFETY STATUS:	"All volunteers accounted for / [Issue: describe and action taken]."
5. NEXT RUN:	"Third run dispatched at 15:00 / No third run — insufficient daylight."

20:00 — END-OF-DAY CHECK-IN

HUB MANAGER CALLS:	"Operations, this is Hub [District Number]. End-of-day check-in. Over."
REPORT FORMAT:	
1. FINAL STOCK COUNT:	"EMK-1: [remaining]. EMK-2: [remaining]. EMK-3: [remaining]. All secured."
2. DAY TOTAL:	"[Number] households reached today. Running total: [cumulative]."
3. INCIDENTS TODAY:	"[No incidents / Incident log entry [ref]: describe and status]."
4. VOLUNTEER STATUS:	"All [number] volunteers returned. [Any injuries, absences]."
5. NEXT-DAY PLAN:	"Tomorrow priority: [zones]. Will need [resources] if conditions allow."

CLOSE:	"Hub [District] end-of-day complete. Standing by on radio overnight. Over and out."
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COORDINATION FAILURE QUICK REFERENCE

Situation	Immediate Action	Escalation if Unresolved
Hub Manager unreachable for >2 hours	Call pre-designated backup contact at sub-warehouse	EC dispatches staff physically to sub-warehouse; EC assumes Hub Manager authority temporarily
Volunteer team >2 hours overdue from delivery run	Hub Manager calls Team Leader, then individual volunteers	After 30 min no contact: notify civil defense with last known route and team size
Local authority withdraws sub-warehouse access	Hub Manager contacts EC immediately	EC calls district-level authority (above ward); Hub Manager activates backup sub-warehouse
Logistics partner fails to deliver trucks	EC contacts backup contractor (second pre-signed agreement)	EC contacts civil defense for vehicle support; stock sub-warehouses sequentially by priority district

SECTION 4 — DELIVERY RUNSHEET TEMPLATE

One per Delivery Run

RUN DETAILS

Sub-Warehouse:	District: _____	Run #:	Date: _____
Team Number:	Team Leader: _____	Depart:	Return: _____
Zone:	Water Depth at Departure: _____ cm	Delivery Mode:	(circle) MOTORBIKE / BICYCLE / BOAT
Volunteers on team:	_____		

HOUSEHOLD DELIVERY LOG

#	Household Name	Address	Score	Band	EMK Type	Qty	Delivered	Signature / Thumbprint	Notes
1							■		
2							■		
3							■		
4							■		
5							■		
6							■		
7							■		
8							■		
9							■		
10							■		
11							■		
12							■		

RUN SUMMARY

Total households reached:	_____	Total EMKs delivered:	_____
EMK-1 delivered:	_____	EMK-2 delivered:	_____
EMK-3 delivered:	_____	New critical cases found:	_____
Route deviations:	_____	Incidents reported:	_____

Team Leader Signature:	_____	Hub Manager Sign-off:	_____
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SECTION 5 — INCIDENT LOG TEMPLATE

Sub-Warehouse Record

SUB-WAREHOUSE DETAILS

District:		Hub Manager:	
Flood Event Dates:		Log Started:	

INCIDENT TYPES

RT — Route Blocked		VS — Volunteer Safety		SS — Stock Scarcity		BF — Building Flooded		OT — Other	
#	Date/Time	Type (code)	Reported By	Description		Status (OPEN/ESC/RESV)	Resolved By	Resolved At	
1						OPEN			
2						OPEN			
3						OPEN			
4						OPEN			
5						OPEN			
6						OPEN			
7						OPEN			
8						OPEN			
9						OPEN			
10						OPEN			

CONTINGENCY DECISIONS LOG

All ad-hoc decisions (route deviations, backup sub-warehouse activation, unplanned collection points) must be logged here.

#	Date/Time	Decision Made	Reason	Authorised By
1				
2				
3				
4				

5				
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SECTION 6 — VOLUNTEER HOUSEHOLD ASSESSMENT FORM

~3 Minutes per Household

INSTRUCTIONS: Complete one form per household. Circle or tick the correct score for each category. Add up the total at the bottom. Return completed forms to Hub Manager every 4 hours. You do NOT need to make a clinical diagnosis — record what you observe and what the household tells you.

Volunteer Name:	_____	Team #:	____	District:	_____
Date:	_____	Time:	____	Zone:	_____
Household Address:				Household ID:	_____
Household Head Name:				Household Size:	_____ persons

CATEGORY 1 — MEDICAL URGENCY

Max 8 points

	Circle or Tick One:	Points
☐	Life-sustaining medication (insulin, heart meds) — run out or less than 24h remaining	8
☐	Chronic illness — medication low (1 to 2 days remaining)	5
☐	Chronic illness — medication currently adequate	2
☐	No chronic illness reported	0
	SCORE FOR THIS CATEGORY:	_____

CATEGORY 2 — HOUSEHOLD VULNERABILITY

Max 5 points — Cap at 5 regardless of combinations

	Circle or Tick One:	Points
☐	Infant under 2 years present	+2
☐	Pregnant woman present	+2
☐	Elderly person 65 or older living alone	+2
☐	Person with physical or cognitive disability present	+2
	SCORE FOR THIS CATEGORY:	_____

CATEGORY 3 — FLOOD EXPOSURE

Max 4 points

	Circle or Tick One:	Points
☐	Water inside household OR building structurally unsafe	4
☐	Water at doorstep (within 10 cm of entrance)	3
☐	Water in street but not reaching household	1
☐	Household is dry and elevated	0
	SCORE FOR THIS CATEGORY:	_____

CATEGORY 4 — SELF-SUFFICIENCY		Max 2 points
	Circle or Tick One:	Points
☐	No clean water, no food, no sanitation access	2
☐	Partial access to some basic necessities	1
☐	Adequate access to basic necessities	0
SCORE FOR THIS CATEGORY:		_____

CATEGORY 5 — ISOLATION		Max 1 point
	Circle or Tick One:	Points
☐	Completely isolated — no neighbors, no family, no phone signal	1
☐	Some social contact or communication available	0
SCORE FOR THIS CATEGORY:		_____

TOTAL SCORE (Add all 5 categories)	_____ / 20
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Score	Priority Band	Delivery Action
15 – 20	CRITICAL	Deliver in current run — goes first
10 – 14	HIGH	Deliver same day
5 – 9	MEDIUM	Deliver within 48 hours
0 – 4	STANDARD	Community collection point self-pickup

Score Band	Recommended EMK Type
CRITICAL or HIGH — Chronic illness (Cat 1 score > 0)	EMK-3 (Chronic Illness Bridge Kit)
CRITICAL or HIGH — Infant, pregnant, elderly, disabled (Cat 2 score > 0)	EMK-2 (Vulnerable Household Kit)
MEDIUM or STANDARD — No special vulnerability	EMK-1 (General Household Kit)

TIEBREAKER RULES (apply in order if two households have the same total score)

Step 1	Higher Category 1 score wins — medical urgency always takes precedence
Step 2	Household with infant under 6 months wins
Step 3	Household whose assessment form was submitted first wins

Volunteer notes (new critical cases, unsafe conditions, refusals, observations):

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Volunteer Signature:	_____	Form Submitted (time):	_____
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